SHEAR: Trimming Spurious Shortcuts from Frozen Text Embeddings via Sparse Autoencoders

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Abstract

Classifiers trained on frozen text embeddings are inexpensive and widely used to label text at scale, but they can learn spurious shortcuts and become biased or prone to distribution shift. If, for example, every randomized trial in the training set studies a neurological disease, they may learn disease area rather than study design. Although existing debiasing methods work well whenever balanced examples or group labels are available, they cannot be used with the biased training set alone. One exception is SHIFT, which ablates sparse autoencoder (SAE) features that encode the attribute, but characterizes them at every hidden layer of the LLM. In this paper we introduce SHEAR (**S**parse **H**ypotheses for **E**mbedding **A**blation and **R**erouting). We show that feature-level debiasing can work across tasks once a single SAE is trained on the task's embedding output rather than at every layer inside an LLM. We train a single SAE on embeddings of a potentially biased training set, use an LLM to label each feature from its explanation, and retrain the classifier with the spurious features either ablated or rerouted by a penalty; neither route needs balanced examples or group labels. Our main results show that the single autoencoder separates the target from the attribute even under perfect correlation in the training data, and that SHEAR surfaces the attribute without being told it. Across seven datasets and three embedding models it raises target accuracy over the unscrubbed classifier and reduces reliance on the shortcut, at a one-time fitting cost and with no LLM in the deployed classifier. We close by probing behaviour under weaker spurious–target correlation rather than claiming broad robustness.

1 Introduction

Classification is regaining a central role in language-model systems: agentic pipelines decompose work into many small judgments (routing a request, screening content, checking whether an action is safe to take) and increasingly delegate them to fast decision models that return a calibrated label in a single pass instead of generating text (Type-Safe AI, 2026). Such models have been branded "System One" models, after the fast, intuitive mode of dual-process accounts of cognition (Kahneman, 2011).

Supervised classifiers trained on frozen text embeddings are the simplest and most transparent way to build such a component: they are cheap to train and run, and their decision rule is a small set of parameters over a fixed representation that can be inspected directly. However, like other neural classifiers, they are vulnerable to spurious correlations; once learned, these shortcuts are systematically applied to all downstream decisions. The problem is especially difficult when the target and spurious attributes are highly correlated: every positive and negative example supports both explanations equally, leaving few counterexamples from which the classifier can learn which signal to rely on. We make the case hardest, and study training data in which the two are *perfectly* correlated.

Many approaches to shortcut mitigation rely precisely on information that is absent in this setting. Methods may require balanced or counterfactually augmented data (Kaushik et al., 2020; Kirichenko et al., 2023), examples in which the target and spurious attributes disagree (Liu et al., 2021a), or explicit labels for the spurious attribute (Ravfogel et al., 2020a; Belrose et al., 2023). When the training data are perfectly correlated, and only target labels are available, these approaches do not apply directly.

An appealing alternative is to intervene in representation space: even when two attributes always co-occur in the training data, they may be encoded along distinct directions, so the classifier's reliance on the spurious directions can be removed without disturbing the target ones.

Sparse Autoencoders (SAEs) decompose neural representations into sparse latent features that can correspond to precisely these interpretable, distinct concepts. Marks et al. (2025) exploit this property in the SHIFT method: features associated with a known spurious concept are identified through natural-language explanations; they are then ablated, reducing a classifier's reliance on that concept without requiring spurious-attribute labels or balanced data.

Given the growing role of lightweight classifiers as decision components in language-model systems, we investigate the possibility of transferring SHIFT's essential insight to classifiers over dense text embeddings. Rather than decomposing internal language-model activations using pretrained SAEs, we train a single SAE directly on the task's text embeddings. This makes the approach substantially simpler and allows the feature vector to be learned from in-domain data. Given only the biased training texts and their target labels, we use the resulting sparse representation to identify features associated with the target and spurious concepts and intervene on the latter. At deployment, the resulting classifier requires neither an LLM nor additional inference-time computation beyond the embedding-based classification pipeline.

Our central question is therefore

> *Without relying on internal language model activations, can a sparse decomposition of dense text embeddings recover and remove a spurious concept when the target and spurious attributes are perfectly correlated?*

We study this question across multiple text-classification settings, examining both whether the learned representation contains separable target and spurious features and whether those features can be identified reliably enough to improve classifier robustness.

We make three contributions.

- We show that a single task-specific SAE over text embeddings recovers target-only and attribute-only features even when the two are perfectly correlated in its training data, and that removing the attribute features improves target accuracy across several dataset families and three embedding models (Section 6).
- We give a label-free recipe for using that recovery: an LLM labels features from their explanations, and a spurious-penalty head reroutes the classifier around them, so debiasing needs neither balanced examples nor group annotations, and adds no LLM to the deployed classifier (Sections 4 and 6).
- We isolate how the recovery works and how far it generalizes: a concept-level association recovers the shortcut from feature meaning, the pipeline surfaces the attribute without being told it on most families, and the shortcut's concentration is an intrinsic property of the dataset and embedding model (Section 6.3).

2 Background

2.1 Shortcut Learning

Let $\mathcal{D} = \{(w_i, y_i)\}_{i=1}^{N}$ denote a dataset of texts $w_i \in \mathcal{W}$ and binary target labels $y_i \in \{0, 1\}$. The goal of supervised text classification is to learn a classifier $\phi : \mathcal{W} \to \{0, 1\}$ that predicts the target label y from the text w.

We assume that each example additionally has a binary spurious attribute $s_i \in \{0, 1\}$ that is correlated with y_i in the training distribution. +

For example, a model trained to classify biomedical abstracts by study type (clinical trial vs. observational study) might inadvertently rely on the disease area (cardiology vs. neurology) if the training data unevenly pairs specific medical subfields with specific experimental designs.

The spurious-attribute labels, s_i, are not observed by the classifier or provided to our method. A classifier exhibits *shortcut reliance* when its predictions depend on this spurious attribute rather than the task-relevant signal associated with the target.

We focus on the challenging setting in which the target and spurious attributes are perfectly correlated in the training data,

$$y_i = s_i \qquad \forall i \in \{1, \ldots, N\}. \tag{1}$$

Consequently, the training data contain no examples in which the target and spurious attributes disagree. Methods that rely on such disagreement to distinguish task-relevant from spurious signals therefore cannot be directly applied in this setting.

2.2 Sparse Autoencoders

Without explicit data-level disagreement, we must isolate and intervene on the classifier's underlying features directly. Sparse Autoencoders (SAEs)

build on classical dictionary learning, which represents high-dimensional observations as sparse combinations of learned basis vectors (Olshausen and Field, 1996, 1997). More recently, the study of superposition in neural networks has motivated the use of dictionary learning to recover interpretable features from neural representations (Elhage et al., 2022; Bricken et al., 2023).

Given an input representation $\mathbf{x} \in \mathbb{R}^d$, an SAE maps $\mathbf{x}$ to a sparse representation $\mathbf{f} \in \mathbb{R}^m$ and reconstructs the original representation as $\hat{\mathbf{x}} \in \mathbb{R}^d$. We refer to each latent dimension f_j as an *SAE feature*.

In our classification adaptation of SHIFT, we use a variant of the TopK SAE architecture introduced by Gao et al. (2025):

$$\mathbf{f} = \mathrm{ReLU}(\mathrm{TopK}(W_{\mathrm{enc}}(\mathbf{x} - \mathbf{b}_{\mathrm{pre}}) + \mathbf{b}_{\mathrm{enc}})),$$
$$\hat{\mathbf{x}} = W_{\mathrm{dec}}\mathbf{f} + \mathbf{b}_{\mathrm{dec}}.$$

The TopK operator retains only the k largest activations, constraining each input to activate at most k SAE features. The SAE is trained to minimize the reconstruction error $\mathcal{L}_{\mathrm{rec}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2$. We additionally use an auxiliary loss to mitigate dead features and follow prior work in the initialization and normalization of the SAE parameters; full training details are provided in Appendix C.

The resulting sparse decomposition allows individual latent features to be inspected and manipulated independently. While prior work has exploited this property to trim spurious features from internal language model activations (Marks et al., 2025), our method adapts this intervention to operate directly on a single, task-specific text embedding bottleneck.

3 Related Work

Shortcut Learning. Neural classifiers are known to exploit spurious attributes that correlate with target labels, rather than learning task-relevant features (Geirhos et al., 2020; Zhou et al., 2024b; McCoy et al., 2019). Existing approaches to mitigating shortcut learning include augmenting the training distribution with counterfactual examples (Kaushik et al., 2020; Zhou et al., 2024a), and retraining the final classification layer on balanced data (Kirichenko et al., 2023). Other methods exploit differences in learning dynamics to identify examples that conflict with a learned shortcut (Liu et al., 2021a).

However, such approaches rely on the presence of examples for which target and spurious attributes disagree.

Representation-level methods instead remove linearly decodable information about an unwanted attribute, typically using attribute labels to identify a subspace to project out (Ravfogel et al., 2020a; Haghighatkhah et al., 2022; Belrose et al., 2023).

SHIFT. In contrast to methods requiring spurious-attribute labels or target–spurious disagreement, Marks et al. (2025) introduced Sparse Human-Interpretable Feature Trimming (SHIFT), which uses Sparse Autoencoders (SAEs) to identify and remove a classifier's reliance on spurious attributes.

As noted, SHIFT decomposes language model activations into sparse latent features, generates natural-language explanations of these features, identifies those associated with a spurious attribute, and ablates them. This procedure reduces a classification head's reliance on gender-related features. Our work builds on SHIFT's central idea of identifying and intervening on interpretable spurious features, but operates on a single SAE trained on the final dense text representation of an embedding model, rather than SAEs over internal language model activations.

Recent work on the "J-space" (Gurnee et al., 2026) likewise operates inside the model, using its Jacobian to identify verbalizable concept representations whose sparsely active contents support downstream computation and can be manipulated to redirect behavior; we ask whether comparable concept-level interventions are available from the output embedding alone.

SAEs on Text Embeddings. While SHIFT and similar methods intervene on the deep internal circuitry of generative models, parallel work has demonstrated that SAEs can similarly decompose static, dense text embeddings into interpretable semantic features.

O'Neill et al. (2024) train SAEs on embeddings of scientific abstracts and demonstrate that the resulting features can be used to interpret and steer semantic search. Movva et al. (2025) use small SAEs to identify interpretable features predictive of a target variable and generate natural-language hypotheses describing them. Kim et al. (2026) intervene on SAE features to remove language-identity information from multilingual embeddings.

This prior work further suggests that training SAEs on domain-specific data can yield features that are more relevant to the domain of interest (O'Neill et al., 2025). Building on these findings, we study whether SAEs over dense text embeddings can mitigate shortcut reliance in text classification. In particular, we consider settings in which the target and spurious attributes are perfectly correlated in the training data, where methods that depend on target–spurious disagreement cannot be directly applied.

4 Method

Our method proceeds in two phases: a diagnostic phase that interprets the classifier in terms of SAE features, and a scrubbing phase that removes the classifier's dependence on the spurious attribute. We call the pipeline SHEAR (Sparse Hypotheses for Embedding Ablation and Rerouting).

4.1 Diagnostic phase

1. **Encode.** We embed every document w via a text embedding model, yielding $\mathbf{x} \in \mathbb{R}^d$.

2. **Decompose.** We train a single TopK sparse autoencoder SAE : $\mathbb{R}^d \to \mathbb{R}^d$ on the biased-set embeddings (Gao et al., 2025; Movva et al., 2025), producing a bottleneck vector $\mathbf{f} \in \mathbb{R}^M$ of M non-negative feature activations, where M is the dictionary width.

3. **Rank.** We rank features by the univariate correlation between their activations and the target label, and keep the top-ranked features for interpretation.

4. **Explain.** For each kept feature, we generate a natural-language explanation from high- and low-activation examples. Practitioners need to inspect these explanations directly.

5. **Label.** We assign each explained feature to one of four classes using one of two channels. The *correlation* channel bins features into *target-only*, *spurious-only*, *mixed*, or *neither* by testing the association between firing patterns and each axis (see Appendix). The *LLM* channel asks an LLM judge to classify the feature explanation into the same four classes from a short description of the target alone, without being told the spurious concept: the judge must infer it.

These two channels capture different notions of a "spurious feature": one behavioral, one semantic. They need not agree, and Section 6.3 measures where they diverge. Because correlation labels require balanced data with spurious annotations, they serve as ground truth for *evaluation only*. The deployed pipeline never sees spurious labels. Feature annotation runs once when the classifier is fitted, and the deployed classifier makes no LLM calls.

4.2 Hard removal by ablation

Hard removal tests the simplest intervention: delete the suspected shortcut directions. Let S be the set of features labeled spurious-only or mixed. We include mixed features because they carry the shortcut even though they also carry target signal; the cost of removing them is measured, not assumed (Section 6.1). Our classifier is a linear head trained on the bottleneck vector with the columns in S zeroed, fitted on the biased training set. This applies the SHIFT operation (Marks et al., 2025; Kim et al., 2026), adapted from LLM internals to a single task-specific SAE over embeddings. Because the head is linear in the features, ablation removes exactly the labeled directions, with no cross-layer attribution and no patching approximation. Everything outside S remains active; the appendix reports a spurious-only variant that ablates less aggressively.

4.3 Soft rerouting by penalty

Ablation struggles when the shortcut and target signals overlap. Under confounding, the shortcut direction also carries the task: nullspace projection collapses target accuracy to chance on three of our four families, and ablating mixed features removes target signal together with spurious signal. The penalty therefore reroutes instead of deleting. It accepts an approximate, label-free semantic S and trains the head to rely on it as little as possible while keeping every feature available. Where masking requires a correct S, the penalty degrades gracefully with a noisy one, which makes the LLM-label pairing viable.

The classifier is one linear layer plus bias, $z = \boldsymbol{\beta} \cdot \mathbf{v} + b$, and its logit splits exactly over the spurious set, $z = z_S + z_{\text{rest}}$. We train with

$$\mathcal{L} = \text{CE} + \lambda R, \qquad R = \frac{\mathbb{E}[z_S^2]}{\mathbb{E}[z^2] + \varepsilon},$$

where R is the fraction of predictive signal flowing through S. We exclude the bias term because the penalty targets the decision direction alone. The

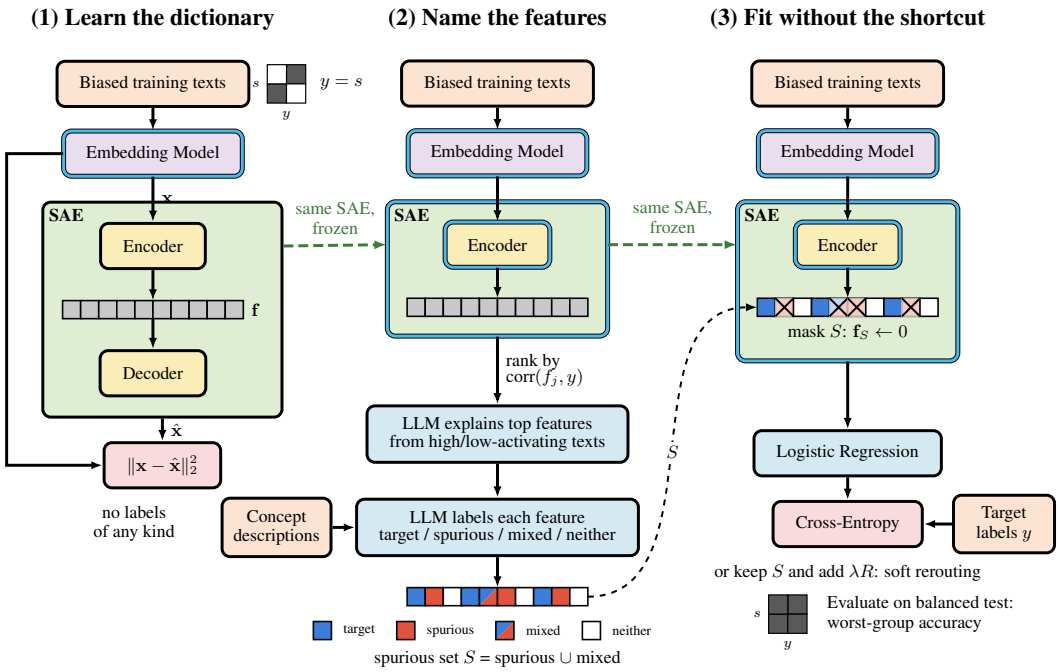


Figure 1: Architecture.

SAE is frozen, so only $\boldsymbol{\beta}$ and the bias move. Setting $\lambda = 0$ reproduces the unpenalized probe and serves as the control. Large λ drives reliance on S toward zero; the cross-entropy term maintains target fit, so λ trades the training proxy R against target accuracy. Success is low R together with high worst-group accuracy on balanced data (Section 6.1). Both normalizations are scale-free, so one λ exerts the same pressure across settings.

We instantiate the penalty in two head spaces. The *bottleneck* head reads SAE activations, and S indexes columns directly. The *embeddings* head reads raw embeddings, while the SAE serves only as the dictionary that names the spurious directions: we mask the batch's activations to S, decode them as $\mathbf{x}_S = W_{\text{dec}}\mathbf{a}_S$, and compute $z_S = \boldsymbol{\beta}\cdot\mathbf{x}_S$. Agreement between the two spaces indicates that the spurious directions are geometrically meaningful in embedding space, not merely coordinate labels. As under ablation, S comes in two flavors: correlation labels or LLM labels.

4.4 Discovery method

The diagnostic phase supports two readings. In the *named-attribute* reading, the practitioner suspects a confounder, supplies its description, and the pipeline returns where it lives in the dictionary (Section 6.3 measures how well). In the *discovery* reading, the practitioner inspects the top-ranked explanations with no attribute in mind and looks for a coherent non-target theme. We test discovery at three levels: *separability*, whether the dictionary contains target-only and spurious-only features despite perfectly correlated training (Section 6.1); *semantic identification*, whether the planted axis is recoverable from meaning alone, either by a concept-association statistic once the concept is hypothesized or by agreement between LLM labels and the behavioural purity classes (Section 6.3); and *open naming*, whether an LLM can name each top-ranked feature's concept *cold*, with no spurious reference, and recover the axis by matching those names (Section 6.4).

4.5 References: skyline and oracle

We use two reference points to bracket both removal methods. The *skyline* runs the same pipeline but selects S with the correlation labels instead of the LLM judge. It measures whether the dictionary contains separable features, isolating dictionary quality from annotator quality: any gap between the skyline and the LLM variant is annotator error, and any gap between the skyline and the oracle is missing separation. The *oracle* trains the same head on the balanced pool. Success means behaving like the oracle while seeing only the biased data.

5 Experiments

5.1 Datasets

We evaluate on seven dataset families (Table 6). Each family consists of a set of documents split twice (balanced and unbalanced).

The biased pool is split into five disjoint sets: `train`, `val`, and `holdout` contain only the two correlated target × spurious cells, so the target label and spurious attribute coincide; `select` and `test` are balanced over all four cells. We split the balanced pool into two disjoint sets, `train` and `val`, to train the oracle. Splits within a pool are disjoint, but the two pools' splits need not be disjoint from each other, so `select` and `test` are shared. We use `select` for diagnostics and SAE selection and reserve `test` for final evaluation of the selected SAE. Split sizes are given in Table 5; see also Appendix B.

Bias in Biographies (De-Arteaga et al., 2019): Professional biographies. TARGET: profession (nurse vs. professor); SPURIOUS: gender.

Pubmed (Sayers et al., 2025): Biomedical abstracts. TARGET: study type (observational vs. trial); SPURIOUS: disease area (cardio vs. neuro).

Amazon Reviews (Hou et al., 2026): Product reviews. TARGET: sentiment (positive vs. negative); SPURIOUS: category (books vs. electronics).

Stack Exchange (Stack Exchange, Inc., n.d.): Stack Overflow posts. TARGET: post type (question vs. answer); SPURIOUS: programming language (Python vs. JavaScript).

Wikipedia (Wikimedia Foundation, 2023; Vrandečić and Krötzsch, 2014; Wikidata contributors, 2026): Encyclopedia articles. TARGET: language edition (English vs. German); SPURIOUS: topic (person vs. creative work). Splits are grouped by Wikidata entity, ensuring no entity overlaps and forming `select`/`test` pairs across editions.

Ulster Newspaper Ads (Jerzak, 2026): Transcribed employment ads from the *Belfast News-Letter* and *Irish News*, 1880–1969. TARGET: occupation (domestic service vs. other); SPURIOUS: religious preference in job candidates mentioned (yes vs. no). The biased pairing (domestic + religion) follows the historical correlation: 30% of domestic-service ads mention religion, against 13% of other types.

Model	Role
`Qwen/Qwen3.5-9B`	Default interpreter + annotator
`Qwen/Qwen3.5-4B`	Annotator-ablation axis
`Qwen/Qwen3.5-0.8B`	Annotator-ablation axis

Table 1: LLMs used.

Global Leadership (Gerring et al., 2024): Political elites, each a synthesized "name, party, country" profile. TARGET: native English speaker vs. not; SPURIOUS: White vs. non-White ethnicity, read from the name. The confound is real but partial ($\rho = .50$ on the natural split), so, as for Ulster, we use the natural split rather than a diagonal-only one: the target and spurious attributes never perfectly coincide. Under it, linear concept erasure (LEACE) holds target accuracy (.90) and removes the shortcut entirely, whereas the perfectly-correlated diagonal split again erases the target.

5.2 Split roles and evaluation protocol

We train the SAE on biased `train` and score reconstruction on `holdout`. We diagnose feature purity on balanced `select`, train scrubbing heads on biased `train`, and evaluate them on balanced `test`. Because `test` is reserved for the single SAE selected per dataset, no comparative claim uses the final evaluation data during selection. Although each pool's splits are disjoint, the balanced pool's `train` and `val`, which train the oracle, may overlap the biased pool's `train`, `val`, and `holdout` by design; no other data are shared.

5.3 Text embedding models

We use the following text embedding models: `OpenAI/GPT-5.4-mini` (closed-source), `Qwen/Qwen3.5-4B` (open-source), `Z.ai/GLM-5.3` (open-source).

5.4 Feature annotation LLMs

We use the LLMs in Table 1 for feature interpretation and annotation. GPT-5.4-mini is the *reference judge* and provides the feature labels behind the main tables; Qwen3.5-4B is the *deployable judge* we ablate over to show a small, locally-served model carries the same protocol; and GLM-5.3 varies the interpreter, which is otherwise fixed across judges.

5.5 SAE configuration and selection

We use the TopK SAE architecture of Gao et al. (2025) as implemented in HypotheSAE (Movva

et al., 2025), training a single SAE over frozen embeddings on the biased `train` split. For each dataset–embedder cell, we sweep sparsity K against dictionary width M, training 52 SAEs per sweep for the original five families (a reduced grid for the two added ones). Because post-removal target accuracy and spurious decodability often trade off, we select two SAEs that bracket the target-accuracy / spurious-decodability Pareto front. **Model A** (best-target) maximizes post-removal target accuracy under the `spurious+mixed` correlation mask. **Model B** (best-spurious) minimizes spurious decodability, excluding degenerate $K{=}2$ cells where near-chance spurious decodability reflects tiny capacity rather than effective scrubbing. We break ties by preferring the smallest M and then the largest K. Where accuracy saturates (Wikipedia on ModernBERT, with approximately 0.999 target accuracy and 0.501 spurious decodability throughout), both picks fall back to the smallest M and largest K by rule. Model A reproduces as the mechanical argmax on 14 of the 15 core cells, with the 15th following the stated tie-break. Model B requires a documented per-cell choice rather than a single derived rule, because the fairness-critical objective is sensitive to degenerate low-capacity solutions. Selection uses only correlation-mask results on balanced `select`; the reserved `test` split and the LLM-labelled arms never enter selection.

We report the two selected SAEs per dataset–embedder cell in Table 7, and anchor all main accuracy results (Table **??**) on these 21 cells. Additional background runs repeat the chosen SAEs across multiple seeds.

5.6 Baselines

We compare against an *oracle* (the same head trained on the balanced pool), the *skyline* (our pipeline with correlation-based feature selection, Section 4), the original *SHIFT* baseline with LLM annotations over Gemma-2-2B internals followed by retraining (Appendix D), and linear concept erasure (*LEACE*; Belrose et al. 2023). Table 2 summarizes each baseline.

5.7 Metrics

All heads are logistic classifiers trained on biased `train` and evaluated on balanced `test`. We report target-mean accuracy, spurious decodability (probe accuracy for the spurious attribute, where chance indicates that the shortcut is no longer readable), and target worst-group accuracy, our primary quantity of interest. Diagnostic claims, including feature purity, explanation fidelity, and annotator agreement, are scored on `select`.

Method
Balanced retraining (oracle) (Idrissi et al., 2022)
Counterfactual augmentation (Kaushik et al., 2020)
Group DRO (Sagawa* et al., 2020)
Just Train Twice (Liu et al., 2021b)
Deep feature reweighting (Kirichenko et al., 2023)
INLP (Ravfogel et al., 2020b) / Mean projection (Haghighatkhah et al., 2022)
LEACE (Belrose et al., 2023) / R-LACE (Ravfogel et al., 2022)
SHIFT (Marks et al., 2025)
Ours
Zero-shot LLM classifier

Table 2: What each method needs beyond the biased training set and its target labels. Concept-erasure methods need supervision for the concept they erase; robust-optimization methods need group annotation; oracle and augmentation methods need balanced (or generated) examples. Feature-scrubbing methods (SHIFT, ours) need only a description of the attribute to remove, and add no inference-time cost over the embedding classifier.

6 Results

Our evaluation tests three claims: first, that scrubbing improves target accuracy on the balanced evaluation set; second, that the dictionary separates the shortcut and a concept-association statistic recovers it; and third, that the pipeline can surface the attribute without being told it. We present them in that order, stress the method along the axes a practitioner would vary, and close with cost.

6.1 Separability: scrubbing improves target accuracy

Table **??** reports mean target accuracy on the balanced `select` split for every (dataset, embedder) cell. Removing the correlation-labelled spurious set (Skyline) raises target accuracy on every family, and by the widest margin exactly where the unscrubbed head is weakest (Amazon Reviews ModernBERT .575→.725; Stack Exchange ModernBERT .597→.807); the balanced-pool Oracle bounds how much is recoverable in principle. The discovery-based LLM mask (SHEAR) recovers most of the skyline gain, matching the correlation mask on some cells (Amazon Reviews Qwen, .837 vs .817) and trailing on others, where the residual mixes annotator error with SAE capacity (Section 6.3). Linear concept erasure (LEACE) collapses to chance on every cell: under perfect corre-

Dataset	Embedder	Target accuracy (balanced `select`)					
		Biased	**SHEAR**	**Skyline**	**Oracle**	**SHIFT**	**LEACE**
Bias in Biographies	Qwen	.766	.834	.961	.956	.926	.500
	ModernBERT	.800	.772	.946	.951	.926	.500
	OpenAI	.721	.760	.961	.959	.926	.500
Pubmed	Qwen	.592	.551	.787	.897	.744	.500
	ModernBERT	.678	.616	.758	.883	.744	.500
	OpenAI	.584	.525	.761	.887	.744	.500
Amazon Reviews	Qwen	.607	.837	.817	.919	.591	.500
	ModernBERT	.575	.711	.725	.887	.591	.500
	OpenAI	.588	.563	.824	.921	.591	.500
Stack Exchange	Qwen	.508	.455	.642	.950	.720	.500
	ModernBERT	.597	.620	.807	.960	.720	.500
	OpenAI	.521	.569	.828	.975	.720	.500
Wikipedia	Qwen	.535	.522	.602	.997	.998	.500
	ModernBERT	.999	.999	.999	1.000	.998	.500
	OpenAI	.646	.585	.782	.998	.998	.500
Ulster Newspaper Ads	Qwen	—	—	—	—	—	.500
	ModernBERT	—	—	—	—	—	.500
	OpenAI	.925	.735	.930	.890	.660	.500
Global Leadership	Qwen	—	—	—	—	—	.500
	ModernBERT	—	—	—	—	—	.500
	OpenAI	.570	.500	.520	.570	—	.500

Table 3: **Full-precision target accuracy (companion to Table ??).** Biased is the unscrubbed head; SHEAR is the discovery (blind, label-free) P2 LLM-label mask (reference judge GPT-5.4-mini; winners-P2 where available else K32/M160); Skyline is the correlation label mask; Oracle is the balanced-pool head; SHIFT is the Gemma-2-2b baseline; LEACE is linear concept erasure. Diagnostic-`select` placeholders pending final `test` re-scores.

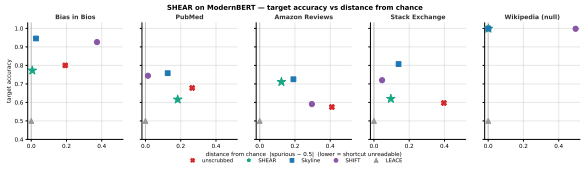


Figure 2: **Debiasing as a two-dimensional trade-off.** Target accuracy against distance from chance, $|S - 0.50|$ (lower is better; a raw sub-chance spurious value would be an *inverted* shortcut, hence we plot distance), on ModernBERT. SHEAR (reference judge) moves every family toward the upper-left, i.e. higher target and a shortcut that is harder to read off; the skyline is the correlation-label ceiling, SHIFT the Gemma-2-2b reference, and LEACE the trivial erasure corner.

lation it removes the target direction together with the spurious one, the failure the penalty is designed to avoid. SHIFT, on its own Gemma-2-2b substrate, raises target accuracy on Bios and PubMed but not on the diffuse families (Appendix 11). Wikipedia×ModernBERT is a near-null-shortcut cell, where the language target is trivially separable, and the topic is not linearly encoded, so we exclude it from the aggregate reading.

6.2 Soft rerouting by penalty

The penalty head (Section 4.3) trains the classifier to route as little of its decision as possible through the spurious set while keeping every feature. At $\lambda = 10$ on the correlation set, it holds Bios target accuracy at the unscrubbed level (.95) while pushing the shortcut toward chance, and on Amazon Reviews, the hardest diffuse shortcut, it keeps target accuracy at the unscrubbed level (.59) with leakage .10. On PubMed and Stack Exchange, it leaves most of the shortcut in place, mirroring the mask. The penalty involves a tradeoff: it trades target accuracy against independence along a tuneable λ frontier rather than deleting features outright, and its full frontier is in the appendix.

6.3 Semantic identification: recovering named shortcuts from explanations

Two channels can label features for removal: a per-feature LLM judge reading each explanation, and the concept-association statistic of Section 4.1. They behave very differently. The concept-association channel recovers the shortcut from feature meaning once the concept is supplied, whereas a lexicon built from the target label re-

covers almost nothing: the shortcut is invisible to target-label statistics, yet the concept-level association reads it off the dictionary. That contrast is the crux of semantic identification: the dictionary separates the shortcut, and the association statistic is the channel that finds it.

6.4 Open-attribute discovery

We test whether the pipeline can identify the planted attribute without being given its name, by naming each top-ranked feature *cold* (the LLM infers the hidden concept from the target and the explanations) and scrubbing with the resulting labels. On most families this raises target accuracy over the unscrubbed head, and on Amazon Reviews it beats the SHIFT baseline; on the remaining families it trails the much larger SHIFT model. Wikipedia is a null-shortcut cell, excluded. Table 4 reports the per-family comparison under the GPT-5.4-mini judge.

6.5 Robustness

Two axes vary widely in our grid. The embedding model spans ModernBERT-large, text-embedding-3-small, and Qwen3-Embedding-4B, whose shortcut geometries differ by up to a factor of two in concentration. The annotator spans GPT-5.4-mini, Qwen3.5-9B, and GLM-5.3, and the qualitative picture above, that association recovers the shortcut while per-feature judging does not, is stable across all three. Seed variation and the weak-correlation sweep are reported in the appendix.

6.6 Cost

The deployed classifier adds no inference-time cost. Annotation runs once when the classifier is fitted; afterward the pipeline is a frozen embedder followed by a linear head, with no LLM in the loop. This is the sharpest contrast with SHIFT, which must decompose and explain features at every layer of an LLM, and it is what makes the method practical for the high-volume decision settings of Section 1. Fitting a single in-domain SAE and its head is a one-time, single-GPU cost.

7 Conclusion

We demonstrate that a single task-specific SAE trained on dense text embeddings can successfully disentangle perfectly correlated spurious shortcuts from task-relevant signals. By rerouting the classifier's predictive signal without requiring balanced data or internal language model activations, this framework offers a scalable, label-free intervention to improve the robustness of lightweight classifiers.

Limitations

Currently, we have no principled way to choose the SAE. We selected the configurations we report using balanced data with spurious labels, information a practitioner would not have.

Also, removal is incomplete. Ablation deletes the features we label, not the attribute itself. Spurious information can survive in features outside the labeled set, absorbed into broader features or split across smaller ones, and a retrained head can find it. The penalty discourages reliance on these features but leaves them in place.

Finally, our empirical setting is specialized: each task has one binary target and one binary spurious attribute, and the correlation is planted by subsampling. Real shortcuts are often partial, multiple, and unnamed; the discovery reading helps with the last only if a person recognizes the theme. We study frozen embeddings with linear heads on seven dataset families, mostly in English.

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	Cold-match accuracy			
Dataset	**Spurious feats → spurious**	**Target feats → target**	**Median margin**	**n named**
Bias in Bios	—	—	—	—
Pubmed	—	—	—	—
Amazon Reviews	—	—	—	—
Stack Exchange	—	—	—	—
Wikipedia Topic	—	—	—	—

Table 4: **Open-attribute discovery (pending cold-naming evaluation).** Each explained feature is named cold and matched to an axis by TF-IDF cosine; agreement with the behavioral purity classes shows the planted axis is recoverable from meanings alone, without ever naming it.

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A Appendix

B Data and split construction

Each source dataset is built in one pass into two pools over the same documents: a *biased* pool and a *balanced* pool. The split sizes are not arbitrary; they derive from a single per-class budget N and five fixed ratios.

Let A, B denote the two *correlated* cells that the biased pool keeps (target = spurious), and C, D the two off-diagonal cells. The biased pool draws

its `train`, `val`, and `holdout` splits from A and B alone, each contributing $1.00N$; its `select` and `test` splits are balanced over all four cells. The per-group fractions are

$$\texttt{train}\,0.50N, \quad \texttt{val}\,0.10N, \quad \texttt{holdout}\,0.10N, \quad \texttt{select}\,0.10N, \quad \texttt{test}\,0.10N,$$

summing to 1.0. The off-diagonal cells therefore fund `select` + `test` + the balanced pool, $0.60N$ each, and the budget is set to the largest value every cell can afford:

$$N \;=\; \min\Big(\min(|A|,|B|),\; \min(|C|,|D|)\,/\,0.60\Big), \tag{2}$$

optionally capped further by a training cap. This spends every correlated-cell record exactly once and every off-diagonal record at most once.

The balanced pool is sized to match the biased pool: it splits into $0.25N$ `train` and $0.05N$ `val` per group, i.e. $1.00N$ and $0.20N$ in total, the same record counts as the biased `train` and `val`. It reuses the biased pool's `select` and `test` files rather than writing its own, and its `train` and `val` may overlap the biased pool's `train`, `val`, and `holdout` by design; every other pair of splits is forced disjoint.

Two modes resolve a rare off-diagonal cell. *Strict* mode (the default) shrinks N until C and D can fund their $0.60N$; *capped* mode instead keeps N at the pair bottleneck and divides the off-diagonal remainder between `select`, `test`, and the balanced pool by weights 1:2:3, sending any unspent correlated surplus back into `train`. Bias in Biographies uses capped mode (the rarest cell, male nurses, holds only 1,735 rows), which is why its `train` exceeds the nominal 50% share.

Wikipedia is the one entity-level source: splitting is done over articles, each of which covers both language editions, and each entity then contributes one record to the biased-only splits and both records to every balanced split, so the balanced `select` and `test` rows form counterfactual pairs. All partitioning uses a fixed seed (42), deduplicates documents before planning, and records the resulting counts in Table 5.

C SAE Training Details

We train our SAEs using the TopK architecture described in Section 2. The number of latent features m and the sparsity level k are varied across experiments.

Auxiliary loss. In addition to the reconstruction objective, we use an AuxK loss to mitigate *dead features*: SAE features that cease to activate and therefore receive no gradient through the standard TopK reconstruction objective. We consider a feature dead if it has not activated during the previous 256 training steps.

The AuxK objective provides an additional gradient path to these dead features. For each input, we select the k_{aux} largest pre-activations among the pool of dead features, denoted by $\mathbf{f}^{\text{AuxK}}$, and use them to reconstruct the residual error left by the main SAE reconstruction, $\mathbf{x} - \hat{\mathbf{x}}$:

$$\mathcal{L}_{\text{AuxK}} = \left\| W_{\text{dec}} \mathbf{f}^{\text{AuxK}} - (\mathbf{x} - \hat{\mathbf{x}}) \right\|_2^2 . \tag{3}$$

Reconstructing the residual rather than the original input encourages dead features to capture information that is not already represented by the currently active features. More importantly, the auxiliary objective allows gradients to reach otherwise inactive features, giving them an opportunity to become active again during subsequent training. Following Movva et al. (2025), we set $k_{\text{aux}} = 2k$.

The complete SAE training objective is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_{\text{aux}} \mathcal{L}_{\text{AuxK}}, \tag{4}$$

where we set $\lambda_{\text{aux}} = 1/32$.

Initialization and normalization. Following the training procedure used by Movva et al. (2025), we:

1. tie the pre-encoder and decoder biases, $\mathbf{b}_{\text{pre}} = \mathbf{b}_{\text{dec}}$, and initialize them to the geometric median of the training embeddings;
2. initialize $W_{\text{enc}} = W_{\text{dec}}^{\top}$; and
3. constrain each decoder feature direction to have unit norm.

Optimization. We optimize the SAE parameters using Adam with a learning rate of 5×10^{-4} and a batch size of 512. We clip gradient norms at 1.0 and train for at most 500 epochs.

When validation embeddings are available, we use early stopping based on the epoch-level validation loss. After each epoch, we evaluate the training objective (Equation 4) over the pre-existing validation split and average the loss across validation batches. We use a patience of five epochs: the patience counter is reset whenever the validation loss strictly improves, and training terminates after five consecutive epochs without improvement.

Dataset	train	val	holdout	select	test		
Bias in Biographies	25,776	3,440	3,440	1,156	2,312	1,974	400
Pubmed	31,680	6,336	6,336	12,672	25,344	21,436	4,280
Amazon Reviews	183,110	36,622	36,622	73,244	146,488	190,238	37,837
Stack Exchange	139,620	27,924	27,924	55,848	111,696	89,440	17,916
Wikipedia	53,856	10,772	10,772	21,544	43,088	53,856	10,772

Table 5: The sizes of the various dataset splits (rows per file). `train`, `val` and `holdout` are perfectly correlated (target = spurious); `select` and `test` are balanced over all four target × spurious groups and shared between the biased and balanced pools.

Dataset	Target Labels	Spurious Labels	Biased Label Pairs
Bias in Biographies	nurse professor	female male	nurse + female professor + male
Pubmed	observational random trial	cardio neuro	observational + cardio random trial + neuro
Amazon Reviews	positive negative	books electronics	positive + books negative + electronics
Stack Exchange	question answer	Python JavaScript	question + Python answer + JavaScript
Wikipedia	English German	person creative work	English + person German + creative work

Table 6: The datasets and their associated target and spurious variable labels. The target and spurious attributes are perfectly correlated in the `train`, `val`and `holdout`splits for each dataset, so they only include data the two listed classes.

D SHIFT Baseline

Gemma 2 2B with Gemmascope 1 SAEs and Neuronpedia annotations

Use all SAEs (residual out, MLP, and attention) for all layers. Did not use the encoder SAE because we didn't have explanations for it.

Use layer 22 residual out for the probe.

Same prompt and model as the other experiments for feature annotation. Used Qwen 3.5 9B as the LLM.

Only remove the top 100 features by signed attribution effect. Not what we want (what the reviewers attacked in the previous paper).

Dataset	Embedder	**Model A** (best-target) K / M	**Target acc.**	**Model B** (best-spurious) K / M	**Spurious acc.**
Bias in Biographies	Qwen	128 / 640	—	32 / 160	—
	ModernBERT	32 / 160	—	32 / 160	—
	OpenAI	128 / 320	—	128 / 320	—
Pubmed	Qwen	32 / 10240	—	8 / 160	—
	ModernBERT	8 / 160	—	8 / 160	—
	OpenAI	32 / 1280	—	8 / 160	—
Amazon Reviews	Qwen	32 / 40960	—	8 / 160	—
	ModernBERT	32 / 640	—	8 / 640	—
	OpenAI	32 / 5120	—	8 / 160	—
Stack Exchange	Qwen	32 / 20480	—	8 / 10240	—
	ModernBERT	32 / 160	—	8 / 320	—
	OpenAI	128 / 5120	—	8 / 2560	—
Wikipedia	Qwen	512 / 2560	—	8 / 40960	—
	ModernBERT	128 / 160	—	128 / 160	—
	OpenAI	2048 / 2560	—	8 / 2560	—

Table 7: Selected SAEs per (dataset $\times$ embedder) cell: the two picks that bracket the target-accuracy / spurious-decodability Pareto front. Model A is the argmax of post-removal target accuracy under the `spurious+mixed` correlation mask; Model B has the lowest spurious decodability excluding degenerate $K{=}2$ cells. Ties break to smallest M, then largest K; the saturated Wikipedia/ModernBERT cell falls back to that rule for both picks. Accuracy cells are filled by the background runs.

		Target accuracy (balanced `select`)					
Dataset	**Embedder**	**Biased**	**SHEAR**	**Skyline**	**Oracle**	**SHIFT**	**LEACE**
Bias in Biographies	Qwen	.766	.834	.961	.956	.926	.500
	ModernBERT	.800	.772	.946	.951	.926	.500
	OpenAI	.721	.760	.961	.959	.926	.500
Pubmed	Qwen	.592	.551	.787	.897	.744	.500
	ModernBERT	.678	.616	.758	.883	.744	.500
	OpenAI	.584	.525	.761	.887	.744	.500
Amazon Reviews	Qwen	.607	.837	.817	.919	.591	.500
	ModernBERT	.575	.711	.725	.887	.591	.500
	OpenAI	.588	.563	.824	.921	.591	.500
Stack Exchange	Qwen	.508	.455	.642	.950	.720	.500
	ModernBERT	.597	.620	.807	.960	.720	.500
	OpenAI	.521	.569	.828	.975	.720	.500
Wikipedia	Qwen	.535	.522	.602	.997	.998	.500
	ModernBERT	.999	.999	.999	1.000	.998	.500
	OpenAI	.646	.585	.782	.998	.998	.500
Ulster Newspaper Ads	Qwen	—	—	—	—	—	.500
	ModernBERT	—	—	—	—	—	.500
	OpenAI	.925	.735	.930	.890	.660	.500
Global Leadership	Qwen	—	—	—	—	—	.500
	ModernBERT	—	—	—	—	—	.500
	OpenAI	.570	.500	.520	.570	—	.500

Table 8: **Full-precision target accuracy (companion to Table ??).** Biased is the unscrubbed head; SHEAR is the discovery (blind, label-free) P2 LLM-label mask (reference judge GPT-5.4-mini; winners-P2 where available else K32/M160); Skyline is the correlation label mask; Oracle is the balanced-pool head; SHIFT is the Gemma-2-2b baseline; LEACE is linear concept erasure. Diagnostic-`select` placeholders pending final `test` re-scores.

		Distance from chance $\lvert S-0.50\rvert$ (balanced `select`)					
Dataset	**Embedder**	**Biased**	**SHEAR**	**Skyline**	**Oracle**	**SHIFT**	**LEACE**
Bias in Biographies	Qwen	.227	.026	.016	.487	.373	.000
	ModernBERT	.193	.006	.027	.489	.373	.000
	OpenAI	.276	.045	.015	.492	.373	.000
Pubmed	Qwen	.389	.202	.125	.472	.014	.000
	ModernBERT	.265	.183	.126	.435	.014	.000
	OpenAI	.398	.077	.164	.466	.014	.000
Amazon Reviews	Qwen	.384	.008	.115	.469	.296	.000
	ModernBERT	.409	.123	.191	.455	.296	.000
	OpenAI	.404	.021	.108	.470	.296	.000
Stack Exchange	Qwen	.483	.059	.202	.490	.046	.000
	ModernBERT	.397	.096	.140	.477	.046	.000
	OpenAI	.475	.173	.146	.487	.046	.000
Wikipedia	Qwen	.465	.368	.398	.497	.496	.000
	ModernBERT	.001	.001	.001	.486	.496	.000
	OpenAI	.354	.301	.218	.499	.496	.000
Ulster Newspaper Ads	Qwen	—	—	—	—	—	.000
	ModernBERT	—	—	—	—	—	.000
	OpenAI	.275	.000	.030	.400	.095	.000
Global Leadership	Qwen	—	—	—	—	—	.000
	ModernBERT	—	—	—	—	—	.000
	OpenAI	.180	.000	.137	.154	—	.000

Table 9: **Full-precision distance from chance $\lvert S-0.50\rvert$ (companion to Table ??).** Lower is better; LEACE reaches 0 only by collapsing the target too. Biased is the unscrubbed head; SHEAR is the discovery (blind, label-free) P2 LLM-label mask (reference judge GPT-5.4-mini; winners-P2 where available else K32/M160); Skyline is the correlation label mask; Oracle is the balanced-pool head; SHIFT is the Gemma-2-2b baseline; LEACE is linear concept erasure. Diagnostic-`select` placeholders pending final `test` re-scores.

		Spurious mean		Target mean		Target worst-group		
Family	#Abl	Pre	Post	Pre	Post	Pre	Post	Δ
Bias in Bios	52	0.739	0.521	0.758	0.950	0.415	0.881	$+0.465$
Stack Exchange	24	0.922	0.763	0.572	0.728	0.069	0.230	$+0.161$
PubMed	42	0.828	0.756	0.654	0.714	0.184	0.277	$+0.093$
McAuley Reviews	24	0.897	0.884	0.586	0.596	0.151	0.126	-0.025
Wikipedia Topic	6	0.502	0.523	0.998	0.977	0.996	0.911	-0.085

Table 10: **K=100** Effect of ablating the spurious features on each dataset family. #Abl is the number of ablated latents; arrows run from pre- to post-ablation. The spurious probe's mean accuracy should fall toward chance, while target worst-group accuracy is the quantity of interest.

		Spurious mean		Target mean		Target worst-group		
Family	#Abl	Pre	Post	Pre	Post	Pre	Post	Δ
Bias in Bios	122	0.739	0.553	0.758	0.926	0.415	0.784	$+0.369$
Stack Exchange	105	0.922	0.766	0.572	0.720	0.069	0.185	$+0.116$
PubMed	153	0.828	0.730	0.654	0.744	0.184	0.474	$+0.291$
McAuley Reviews	163	0.897	0.886	0.586	0.591	0.151	0.205	$+0.054$
Wikipedia Topic	55	0.502	0.502	0.998	0.998	0.996	0.996	$+0.000$

Table 11: **k=1000** Effect of ablating the spurious features on each dataset family. #Abl is the number of ablated latents; arrows run from pre- to post-ablation. The spurious probe's mean accuracy should fall toward chance, while target worst-group accuracy is the quantity of interest.